

SACHET KORADA

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EDUCATION

Bachelor of Science in Robotics, Minor in Business Administration
Carnegie Mellon University, GPA: 3.65/4.00

Expected Graduation: 2028

Relevant Coursework: Computer Systems, Computer Vision, General Robotics, Feedback Control Systems, Sensing & Sensors, Foundational Mathematics of Robotics, Robotics & AI in Agriculture, Multi-Robot Planning & Coordination

SKILLS

Computer Languages	C++, Python, Java, MATLAB
Libraries & Frameworks	OpenCV, YOLO, ROS 2, Open3D, CUDA, PyTorch, Google Protobuf, Docker
Sensors & Robotics Hardware	LiDAR, RGB-D Cameras, INS, ESP-32, Raspberry Pi, NVIDIA Jetson

EXPERIENCE

Perception Engineering Intern
Anduril Industries

May – August 2026
Costa Mesa, CA

- Developed **multi-frame bundle adjustment** for **EO/LWIR-to-LiDAR extrinsic calibration**, jointly optimizing camera fiducials and LiDAR point-to-board residuals to achieve **5.1 mm median** LiDAR error and **0.24 px median** reprojection error and accelerated the overall pipeline by **211×** through vectorization and parallelization.
- Designed a continuous-time **7+ sensor calibration** system using SfM, LiDAR correspondences, cubic B-spline trajectories, robust estimation, and optimization on SE(3), achieving **0.001° / 1.6 mm** gauge-free camera-to-camera relative-pose accuracy (**30× better than target**).
- Implemented a **C++/Ceres fixed-lag information smoother** with analytic Jacobians, Schur-complement landmark marginalization, and covariance estimation; diagnosed calibration gauge degeneracies through Fisher-information analysis.

Software Engineering Intern
Hughes Networking Systems

June – August 2025
Gaithersburg, MD

- Developed **C++ simulation software** for large-scale satellite network stress testing, modeling **100+** concurrent users interacting with the Anchor Control Processor through **socket programming** and **Google Protobuf**.
- Automated validation workflows with **gTest/gMock**, creating **20+** unit and integration tests for concurrent network components while identifying **race conditions** and reliability bottlenecks.

RESEARCH

Tree of Heaven Detection System

August 2025 – May 2026

Second author, manuscript submitted to IEEE Robotics and Automation Letters (RA-L)

- Built **geospatial maps** of invasive Tree-of-Heaven locations by **aligning aerial imagery** with **LiDAR point clouds** using **photogrammetry** and fusing **YOLOv11 segmentation** outputs with **3D spatial data**.
- Trained **YOLO segmentation models** on **1,000+ labeled images**, achieving **0.71 mAP** and enabling automated detection of invasive Tree-of-Heaven plants in field imagery
- Accelerated large-scale inference by **optimizing GPU pre-processing** and batch inference pipelines using **CUDA-enabled PyTorch**.

PROJECTS

CMU RoboBuggy — Multi-Sensor Perception System

August 2025 – Present

- Architected a **real-time** LiDAR-camera sensor fusion pipeline in **ROS 2** for an autonomous racing vehicle, associating camera detections with projected 3D LiDAR clusters at **30 Hz** across **30+ recorded driving logs**.
- Implemented **RANSAC-based ground segmentation** and obstacle classification to extract structured features from noisy **3D point-cloud** data.
- Developed **visualization** and **debugging tools** to analyze perception outputs across recorded datasets and live field tests, **improving error analysis** and model validation workflows.